

# Aflevering P5

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## Spørgsmål 1

Forhold dig til den feedback du fik til aflevering P4

Jeg fik at vide at min animation ikke helt var som den skulle være, og at jeg skulle undgå at bruge for og if loops i update-funktionen

## Øgleopgave

Vi vender tilbage til MC Escher. Han er bl.a. kendt for sin "lizard-tegning":



Der findes en interaktiv hjemmeside, der kan hjælpe dig med at gennemskue hvordan mønsteret opstår.

Den er her: <https://www.geogebra.org/m/zxuaxtk8>

## Spørgsmål 4

Færdiggør denne klasse så den kan benyttes til at få tegnet en øgle et vilkårligt sted, roteret en vilkårlig vinkel og med en vilkårlig farve:

```
class Lizard():
```

```
    xs = [ -53, -49, -46, -43, -39, -36, -33, -28, -23, -17, -11,
           -6,  0,  4,  2,  1, -1, -2, -3, -7, -10, -14,
          -18, -22, -18, -11, -3,  6, 14, 18, 21, 24, 26,
           29, 32, 35, 38, 41, 49, 58, 68, 78, 87, 97,
          106, 110, 114, 118, 122, 125, 119, 113, 107, 100, 94,
           88, 81, 76, 69, 60, 57, 56, 55, 55, 58, 63,
           71, 78, 85, 92, 100, 107, 115, 120, 126, 131, 137,
          143, 148, 154, 160, 160, 159, 154, 149, 144, 139, 134,
          129, 124, 119, 114, 105, 97, 89, 81, 73, 65, 57,
           50, 43, 38, 38, 37, 36, 35, 34, 36, 41, 47,
           52, 58, 63, 69, 74, 76, 72, 69, 62, 54, 47,
           40, 32, 27, 24, 22, 24, 23, 16, 10, 3, -4,
          -11, -10, -9, -7, -6, -5, -4, -3, -7, -12, -16,
          -21, -26, -27, -25, -23, -21, -24, -27, -31, -34, -38,
          -41, -46, -51, -56, -61, -68, -75, -82, -89, -96, -104,
```

```

-112, -120, -126, -131, -137, -142, -143, -139, -134, -129, -124,
-119, -114, -106, -98, -91, -83, -75, -68, -66, -64, -65,
-70, -75, -79, -83, -85, -86, -87, -89, -97, -105, -114,
-124, -133, -142, -151, -156, -161, -166, -172, -177, -182, -188,
-186, -184, -181, -178, -176, -173, -170, -165, -158, -151, -144,
-137, -130, -123, -115, -106, -97, -89, -81, -76, -72, -68,
-64, -60, -56, -52, -48, -48, -51, -56, -60, -65, -70,
-74, -79, -84, -88, -91, -86, -81, -76, -72, -67, -62,
-57, -53]
ys = [-200, -193, -187, -180, -173, -167, -160, -154, -150, -145, -141,
-137, -133, -127, -119, -110, -102, -93, -84, -78, -71, -65,
-59, -53, -48, -45, -43, -42, -40, -46, -53, -60, -68,
-75, -82, -89, -96, -103, -103, -102, -102, -102, -101, -101,
-100, -95, -89, -83, -77, -70, -66, -62, -58, -55, -51,
-47, -48, -53, -57, -57, -50, -41, -32, -23, -17, -13,
-15, -18, -21, -24, -26, -29, -31, -27, -22, -18, -14,
-9, -5, 0, 4, 13, 22, 27, 32, 37, 42, 47,
52, 57, 62, 67, 66, 64, 62, 60, 58, 56, 54,
53, 56, 61, 71, 80, 89, 98, 107, 114, 118, 123,
127, 132, 136, 141, 145, 151, 157, 164, 167, 169, 172,
175, 177, 173, 166, 158, 150, 144, 140, 137, 134, 131,
127, 118, 109, 101, 92, 83, 74, 65, 60, 54, 49,
43, 39, 45, 52, 60, 68, 75, 82, 88, 95, 101,
108, 113, 118, 123, 128, 125, 122, 119, 116, 113, 115,
117, 119, 117, 112, 107, 102, 97, 91, 86, 81, 76,
71, 66, 66, 68, 70, 73, 75, 74, 66, 58, 51,
46, 41, 36, 30, 21, 12, 3, -4, -6, -8, -9,
-9, -8, -7, -6, -1, 4, 9, 13, 18, 23, 26,
18, 10, 3, -4, -12, -19, -26, -31, -34, -37, -40,
-43, -46, -49, -51, -52, -53, -55, -55, -58, -64, -70,
-76, -82, -88, -94, -100, -105, -111, -116, -121, -127, -132,
-137, -143, -148, -153, -159, -164, -169, -174, -180, -185, -190,
-195, -200]

def __init__(self, ax, center=[0,0], rotation_in_deg=0, farve='C0'):
    self.center =
    self.rotation =
    self.polygon = ax.fill

def tegn(self):
    R =
    P = np.dot(R,np.row_stack((
    P[0] +=
    P[1] +=
    self.polygon.set_xy(P.T)

```

```

In [1]: import numpy as np
import matplotlib.pyplot as plt

plt.rc("xtick", labels=16)
plt.rc("ytick", labels=16)
plt.rc("axes", titles=20)
plt.rc("font", size=16)

fig, ax = plt.subplots()

ax.set_aspect('equal')
ax.grid('on')
ax.xaxis.set_major_locator(plt.MultipleLocator(200))
ax.yaxis.set_major_locator(plt.MultipleLocator(200))
ax.set_xlim([-10, 1200])
ax.set_ylim([-10, 800])

```

```
plt.close()
```

```
class Lizard():
```

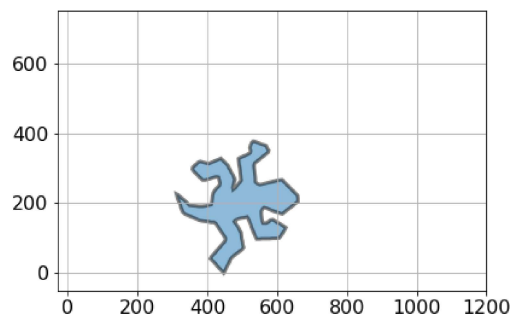
```
    xs = [ -53,  -49,  -46,  -43,  -39,  -36,  -33,  -28,  -23,  -17,  -11,
           -6,   0,   4,   2,   1,  -1,  -2,  -3,  -7, -10, -14,
          -18, -22, -18, -11,  -3,   6,  14,  18,  21,  24,  26,
           29,  32,  35,  38,  41,  49,  58,  68,  78,  87,  97,
          106, 110, 114, 118, 122, 125, 119, 113, 107, 100, 94,
           88,  81,  76,  69,  60,  57,  56,  55,  55,  58,  63,
           71,  78,  85,  92, 100, 107, 115, 120, 126, 131, 137,
          143, 148, 154, 160, 160, 159, 154, 149, 144, 139, 134,
          129, 124, 119, 114, 105, 97, 89, 81, 73, 65, 57,
           50,  43,  38,  38,  37,  36,  35,  34,  36,  41,  47,
           52,  58,  63,  69,  74,  76,  72,  69,  62,  54,  47,
           40,  32,  27,  24,  22,  24,  23,  16,  10,   3,  -4,
          -11, -10,  -9,  -7,  -6,  -5,  -4,  -3,  -7, -12, -16,
          -21, -26, -27, -25, -23, -21, -24, -27, -31, -34, -38,
          -41, -46, -51, -56, -61, -68, -75, -82, -89, -96, -104,
          -112, -120, -126, -131, -137, -142, -143, -139, -134, -129, -124,
          -119, -114, -106, -98, -91, -83, -75, -68, -66, -64, -65,
          -70, -75, -79, -83, -85, -86, -87, -89, -97, -105, -114,
          -124, -133, -142, -151, -156, -161, -166, -172, -177, -182, -188,
          -186, -184, -181, -178, -176, -173, -170, -165, -158, -151, -144,
          -137, -130, -123, -115, -106, -97, -89, -81, -76, -72, -68,
          -64, -60, -56, -52, -48, -48, -51, -56, -60, -65, -70,
          -74, -79, -84, -88, -91, -86, -81, -76, -72, -67, -62,
          -57, -53]
```

```
    ys = [-200, -193, -187, -180, -173, -167, -160, -154, -150, -145, -141,
          -137, -133, -127, -119, -110, -102, -93, -84, -78, -71, -65,
          -59, -53, -48, -45, -43, -42, -40, -46, -53, -60, -68,
          -75, -82, -89, -96, -103, -103, -102, -102, -102, -101, -101,
          -100, -95, -89, -83, -77, -70, -66, -62, -58, -55, -51,
          -47, -48, -53, -57, -57, -50, -41, -32, -23, -17, -13,
          -15, -18, -21, -24, -26, -29, -31, -27, -22, -18, -14,
           -9,  -5,   0,   4,  13,  22,  27,  32,  37,  42,  47,
           52,  57,  62,  67,  66,  64,  62,  60,  58,  56,  54,
           53,  56,  61,  71,  80,  89,  98, 107, 114, 118, 123,
          127, 132, 136, 141, 145, 151, 157, 164, 167, 169, 172,
          175, 177, 173, 166, 158, 150, 144, 140, 137, 134, 131,
          127, 118, 109, 101, 92, 83, 74, 65, 60, 54, 49,
           43,  39,  45,  52,  60,  68,  75,  82,  88,  95, 101,
          108, 113, 118, 123, 128, 125, 122, 119, 116, 113, 115,
          117, 119, 117, 112, 107, 102, 97, 91, 86, 81, 76,
           71,  66,  66,  68,  70,  73,  75,  74,  66,  58,  51,
           46,  41,  36,  30,  21,  12,   3,  -4,  -6,  -8,  -9,
           -9,  -8,  -7,  -6,  -1,   4,   9,  13,  18,  23,  26,
           18,  10,   3,  -4, -12, -19, -26, -31, -34, -37, -40,
          -43, -46, -49, -51, -52, -53, -55, -55, -58, -64, -70,
          -76, -82, -88, -94, -100, -105, -111, -116, -121, -127, -132,
          -137, -143, -148, -153, -159, -164, -169, -174, -180, -185, -190,
          -195, -200]
```

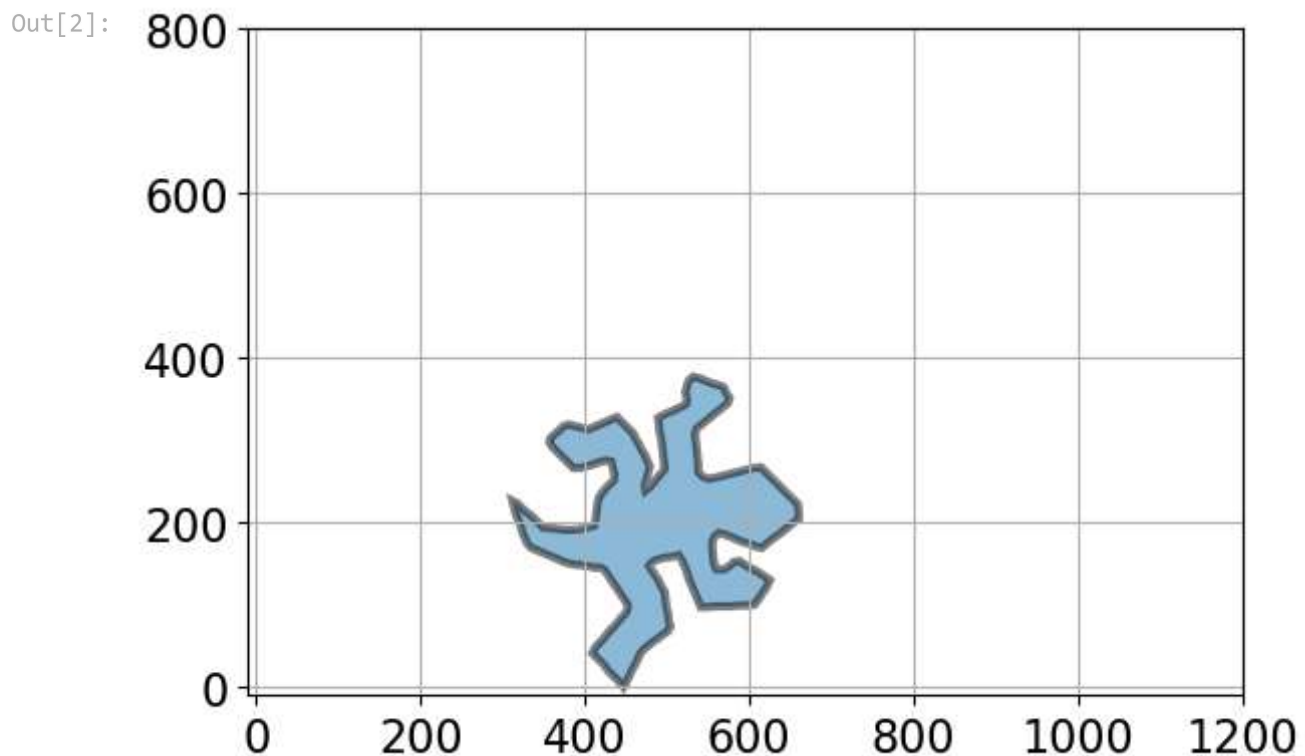
```
    def __init__(self, ax, center=[0,0], rotation_in_deg=0, farve='C0'):
        self.center = np.array(center)
        self.rotation = rotation_in_deg*np.pi/180 #Til Radianer
        self.polygon = ax.fill([],[],farve, alpha=0.5, ec="k", lw=3)[0]
```

```
    def tegn(self):
        R = np.array([[np.cos(self.rotation), -np.sin(self.rotation)],
                      [np.sin(self.rotation), np.cos(self.rotation)]])
        P = np.dot(R,np.row_stack((self.xs,self.ys)))
        P[0] += self.center[0]
        P[1] += self.center[1]
        self.polygon.set_xy(P.T)
```

Lav en instans af din klasse og få den til at tegne sig selv.

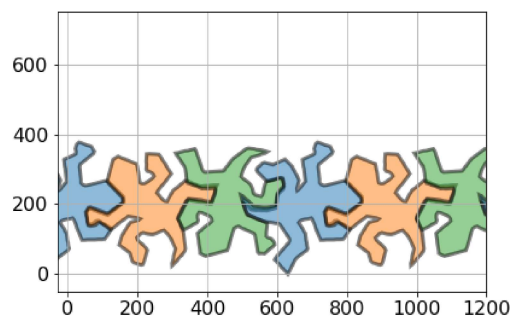


```
In [2]: lizard1 = Lizard(ax, [500, 200], 0, 'C0')
lizard1.tegn()
fig
```



## Spørgsmål 6

Lav mange instanser af din klasse idet du flytter dem 228 enheder til højre og drejer dem  $120^\circ$  hver gang du laver en ny. Benyt en `for`-løkke. Plot det hele.



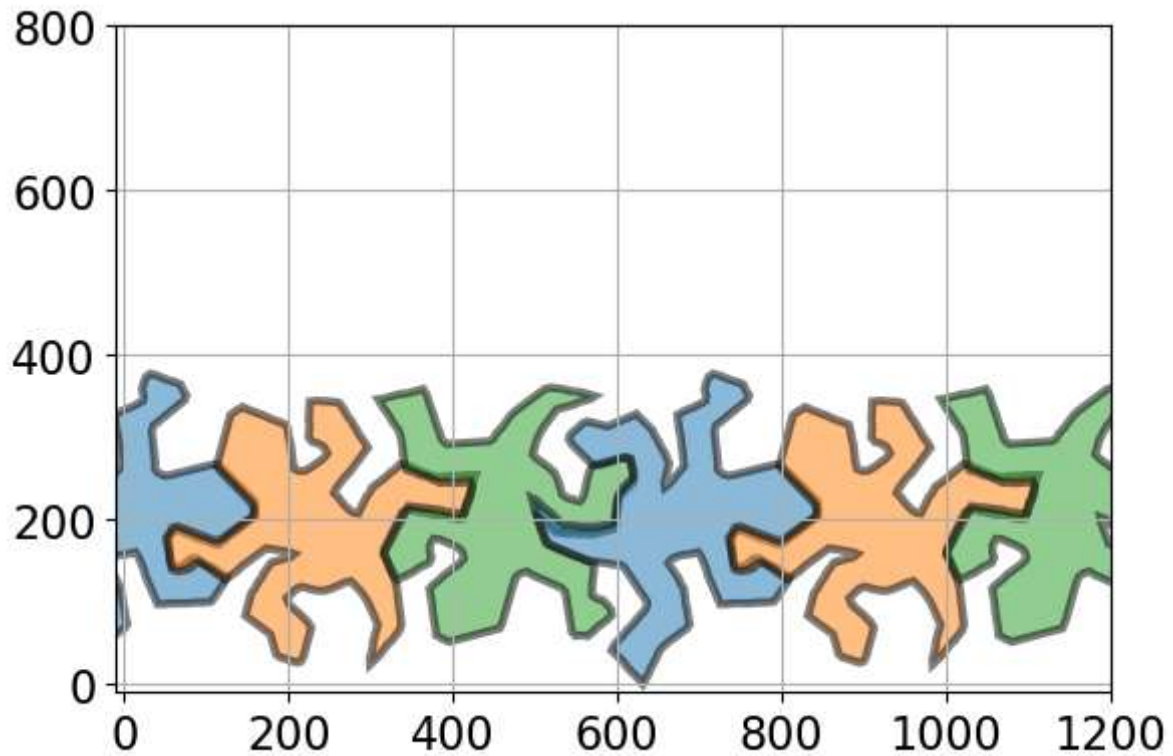
```
In [3]: fig, ax = plt.subplots()

ax.set_aspect('equal')
ax.grid('on')
ax.xaxis.set_major_locator(plt.MultipleLocator(200))
ax.yaxis.set_major_locator(plt.MultipleLocator(200))
ax.set_xlim([-10, 1200])
```

```
ax.set_ylim([-10, 800])

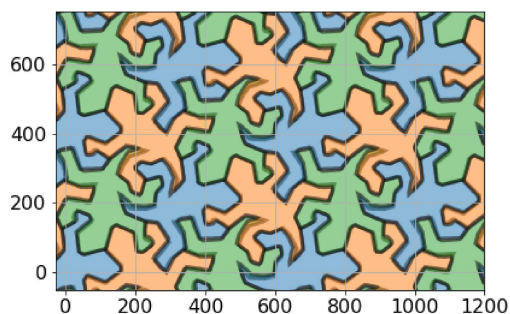
centrum = [0,200]
rot = 0
farver = ['C0', 'C1', 'C2']

for i in range(6):
    farve = farver[i%3]
    liz = Lizard(ax,centrum,rot, farve)
    liz.tegn()
    centrum += np.array([228,0])
    rot += 120
```



## Spørgsmål 7

Fyld hele fladen ud med rækker af øgler. Hver ny række skal flyttes "noget" til en side og 200 enheder op. Benyt endnu en `for`-løkke. Plot det hele.



```
In [4]: fig, ax = plt.subplots()

ax.set_aspect('equal')
ax.grid('on')
ax.xaxis.set_major_locator(plt.MultipleLocator(200))
ax.yaxis.set_major_locator(plt.MultipleLocator(200))
ax.set_xlim([-20, 1200])
ax.set_ylim([-20, 800])

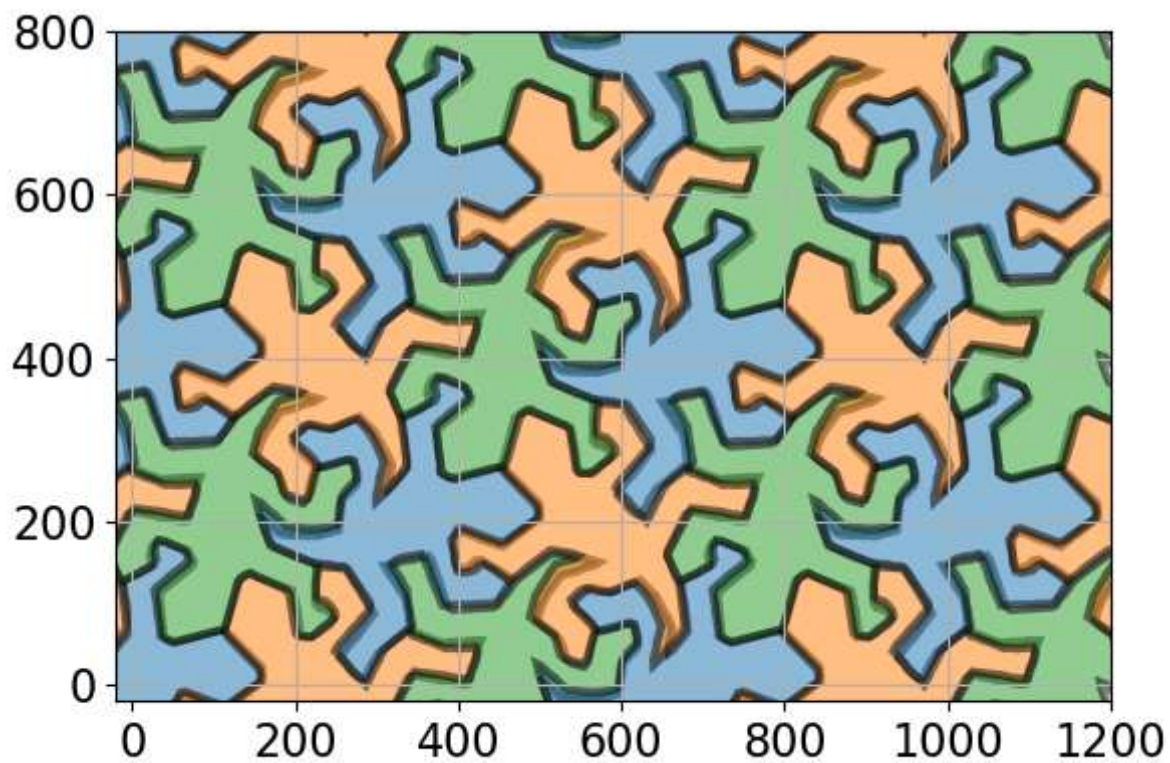
centrum = [0,0]
rot = 0
```

```

farver = ['C0', 'C1', 'C2']
noget = 340

for j in range(-2,6):
    for i in range(-3,6):
        centrum = [i*228,j*200]
        if j%2 != 0:
            centrum = [noget + i*228,j*200]
        rot = i*120
        farve = farver[i%3]
        liz = Lizard(ax,centrum,rot, farve)
        liz.tegn()

```



In [ ]: